MATH 350-2 Advanced Calculus

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Outline

1. Real Analysis Lecture 20
   - More on differentiation

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Local maxima and minima

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It has **local maximum at** $a \in S$ if there exists an $r > 0$ such that

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Absolute and local minima are defined similarly.

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Theorem

Let f be a real-valued function on an open interval (a, b) and suppose that f has a local maximum or local minimum at $c \in (a, b)$. Then c is a critical point of f.

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To obtain a contradiction, suppose that $f'(c) \neq 0$.

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To obtain a contradiction, suppose that $f'(c) \neq 0$.
Then by the Helper Theorem, there is a function $f^*(x)$ defined on (a, b) and continuous at $c \in (a, b)$ with

$$f(x) - f(c) = (x - c)f^*(x).$$



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It follows that $f^*(x) > |f^*(c)|/2$ for all $x \in (a, b)$ with $|x - c| < \delta$.

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It follows that $f^*(x) > |f^*(c)|/2$ for all $x \in (a, b)$ with $|x - c| < \delta$.
However,

$$f(c \pm \delta/2) = f(c) \pm \frac{1}{2}\delta f^*(c + \delta/2),$$

one of these will be larger than $f(c)$.

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However,

$$f(c \pm \delta/2) = f(c) \pm \frac{1}{2}\delta f^*(c + \delta/2),$$

one of these will be larger than $f(c)$. Contradiction! □

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Then there exists $c \in (a, b)$ with $f'(c) = 0$.

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Then $f'(c) = 0$ for every $c \in (a, b)$.

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Then $f'(c) = 0$ for every $c \in (a, b)$.
Otherwise at least one of the points happens inside, so $f(x)$ has a local max or local min at a value c inside (a, b).

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Then $f'(c) = 0$ for every $c \in (a, b)$.
Otherwise at least one of the points happens inside, so $f(x)$ has a local max or local min at a value c inside (a, b).
The point c is a critical point, and since f is differentiable at c, we must have $f'(c) = 0$. □

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Then there exists $c \in (a, b)$ with

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Mean Value Theorem

Proof.

Consider the function

$$g(x) = f(x) - \left(\frac{f(b) - f(a)}{b - a} \right) (x - a) + f(a).$$

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Then $g(x)$ is

- continuous on $[a, b]$;
- differentiable at each point in (a, b);
- and satisfies $g(a) = g(b)$.

By Rolle's Theorem, there exists $c \in (a, b)$ with

$$0 = g'(c) = f'(c) - \frac{f(b) - f(a)}{b - a}.$$

